IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS 1A

Wednesday, 8 January 2025, 10:00 AM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

This exam has TWO sections.
Section A (Question ONE) is worth 40 Marks, and is multiple choice, please enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks.
Section B has THREE longer questions, worth 20 marks each. Enter your answers in the booklet provided.

NO CALCULATORS ALLOWED. Formula sheet provided

Any special instructions for invigilators and information for candidates are on page 1.

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ELEC40012 - MATHEMATICS1A - QUESTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet, A. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex $z = x + iy$, which of the following are the real and imaginary parts of $w = \cos(iz^2)$?

   (a) $\mathrm{Re}(w) = -\cosh(2xy)\cos(x^2-y^2)$, and $\mathrm{Im}(w) = \sinh(2xy)\sin(x^2-y^2)$,

   (b) $\mathrm{Re}(w) = \cosh(2xy)\sin(x^2-y^2)$, and $\mathrm{Im}(w) = \sinh(2xy)\cos(x^2-y^2)$,

   (c) $\mathrm{Re}(w) = \sin(2xy)\cosh(x^2-y^2)$, and $\mathrm{Im}(w) = -\cos(2xy)\sinh(x^2-y^2)$,

   (d) $\mathrm{Re}(w) = \cos(2xy)\cosh(x^2-y^2)$, and $\mathrm{Im}(w) = \sin(2xy)\sinh(x^2-y^2)$,

   (e) none of the above. [3]

2. If n is any integer, which of the following is the locus $z = x + iy$, in the complex plane, satisfying $Arg(z-i) = \dfrac{\pi}{4} + n\pi$?

   (a) the line $y = x+1$, excluding $(0,1)$, (b) the line $y = -x-1$, excluding $(-1,0)$,

   (c) the line $y = -x+1$, excluding $(1,0)$, (d) the line $y = x-1$, excluding $(0,-1)$,

   (e) none of the above. [3]

3. Which of the following statements about $\lim_{x\to\infty} x^q\left(\sqrt{x-2}-\sqrt{x+2}\right)$ is true?

   (a) If $q = 1$, the limit is equal to 0, (b) If $q = -\dfrac{1}{2}$, the limit does not exist,

   (c) If $q = \dfrac{1}{2}$, the limit is equal to 2, (d) the limit cannot be determined,

   (e) none of the above. [3]

4. The limit given as $\lim_{x\to\frac{\pi}{2}}\left(\dfrac{1-\sin x}{\frac{\pi}{2}-x-\cos x}\right)$

   (a) exists, but cannot be determined, (b) does not exist,

   (c) is equal to -2, (d) is equal to 0,

   (e) none of the above. [3]

5. The motion of a particle at time $t > 0$, is described by x, with velocity $\dfrac{dx}{dt} = \dfrac{\ln t}{t}$. At time t_1, the particle achieves maximal velocity, and at time t_2, the curvature of the velocity graph changes. Hence

   (a) $t_1 = 0$ and $t_2 = e$, (b) $t_2 = e$ and $t_1 = e^{2/3}$,

   (c) $t_1 = e^{3/2}$ and $t_2 = e$, (d) $t_2 = e$ and $t_1 = 1$,

   (e) none of the above. [2]

6. If $\cos x - \sin y = e^{x+y}$, which of the following is $\dfrac{dy}{dx}$?

   (a) $-\dfrac{e^{x+y} - \sin y}{\cos x - e^{x+y}}$, (b) $-\dfrac{e^{x+y} + \cos x}{\sin y - e^{x+y}}$, (c) $-\dfrac{e^{x+y} + \cos y}{\sin x - e^{x+y}}$, (d) $-\dfrac{e^{x+y} + \sin x}{\cos y + e^{x+y}}$,

   (e) none of the above. [2]

7. If $y = \sin v$, where $v = u + \dfrac{1}{u}$ and $u = \ln x$, what is the value of $\dfrac{dy}{dx}$ at $x = e$?

   (a) $\dfrac{1}{e}$, (b) -1 , (c) $\dfrac{2}{e^2}$, (d) $\sec^2(e)$,

   (e) none of the above. [2]

8. If $I(t) = \displaystyle\int_0^t x^2 e^{-2x}\, dx$ is a function of t, which of the following statements is **true**?

   (a) $I(t)$ has a local maximum at $t = 1$, (b) $I(t)$ has a point of inflection at $t = 1$,

   (c) $I(t)$ is decreasing for $t > 1$, (d) $I(t)$ has a point of inflection at $t = 1 + \dfrac{1}{\sqrt{2}}$,

   (e) none of the above . [3]

9. The integral $I = \displaystyle\int \frac{1}{1 + \sin x}\, dx$ is equal to

   (a) $-\dfrac{2}{1 + \tan(x/2)} + \mathrm{C}$, (b) $\tan x + \sec x + \mathrm{C}$, (c) $\dfrac{x}{1 + \sin x} + \mathrm{C}$, (d) $\dfrac{2}{\cot(x/2) - 1} + \mathrm{C}$,

   (e) none of the above. [3]

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}} .$$

    Which of the following statements is true?

    (a) S_1 diverges for $\varepsilon \geq 0$, and S_2 has radius of convergence $R = 1/2$,

    (b) S_1 converges for $\varepsilon \leq 0$, and S_2 has radius of convergence $R = 1$,

    (c) S_1 diverges for $\varepsilon \leq 0$, and S_2 has interval of convergence $[-1, 1)$,

    (d) S_1 has radius of convergence $R = 1$, and S_2 diverges for $|x| > 1$,

    (e) none of the above. [3]

11. What is wrong with the following argument about the Maclaurin series for $f(x) = \sqrt{1+x}$?

To first order with remainder term, the series is given by $f(x) = 1 + \frac{1}{2}x + R_1$, *(1)*

where $R_1 = \frac{f''(c)}{2}x^2 = \frac{1}{8(1+c)^{3/2}}x^2$, *with c between 0 and x.* *(2)*

If we estimate $\sqrt{0.9} \approx 1 + \frac{1}{2}(-0.1) = 0.95$, *(3)*

the remainder can be used to give a bound for the maximum error:

$$|E| < \frac{1}{800(0.9)^{3/2}} \, . \qquad (4)$$

(a) the coefficients of the series in (1) are wrong,

(b) the coefficient of the remainder in (2) is wrong ,

(c) the errror in (4) is bounded using the wrong value of c,

(d) there is nothing wrong with the argument ,

(e) none of the above. [3]

12. To solve the linear first order ordinary differential equation (ODE)

$$\sec(x)\frac{dy}{dx} - \frac{2y}{\sin(x)} = 1$$

we obtain the integrating factor $\mu(x)$:

(a) $\mu(x) = \csc^2(x)$, (b) $\mu(x) = \ln[\tan(x/2)]$, (c) $\mu(x) = -\sin^2 x$, (d) $\mu(x) = e^{-\int \frac{2}{\sin x} dx}$,

(e) none of the above . [2]

13. To solve the linear first order ODE

$$\sec(x)\frac{dy}{dx} - \frac{2y}{\sin(x)} = 1 \, ,$$

satisfying the initial condition $y(\pi/2) = 1$, we use the integrating factor $\mu(x)$, found in the previous question, to obtain

(a) $y = \cos x(2\sin x + 1)$, (b) $y = \sin x(2\sin x - 1)$,

(c) $y = \frac{1}{\cos x}(1 - 2\sin x)$, (d) $y = \frac{1}{\sin x}(2 + \cos x)$,

(e) none of the above. [3]

14. The subtstitution $x = e^t$ transforms the Euler equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0\,,$$

into which of the following linear second-order ODEs?

(a) $\dfrac{d^2y}{dt^2} - 2\dfrac{dy}{dt} + y = 0\,,$ (b) $\dfrac{d^2y}{dt^2} + 3\dfrac{dy}{dt} + y = 0,$

(c) $\dfrac{d^2y}{dt^2} - 3\dfrac{dy}{dt} + y = 0\,,$ (d) $\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} - y = 0\,,$

(e) none of the above. [2]

15. After employing the substitution in the previous question, the solution of the Euler equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0\,,$$

is found to be

(a) $y = \dfrac{1}{x}(1+\ln x) + C\,,$ (b) $y = \dfrac{x}{1+\ln x} + C\,,$

(c) $y = x(1+\ln x) + C\,,$ (d) $y = \dfrac{1}{x}(\ln x - 1) + C\,,$

(e) none of the above. [3]

Section B

Enter your full answers in the answer booklet, B. This section has three questions, each worth 20 marks.

1. a) Solve the equation $\cosh x - \frac{3}{2}\sinh x = 1$, for real x.
Hence solve $\cosh z - \frac{3}{2}\sinh z = 1$, for complex z.
Hence, or otherwise, solve $\cos x - \frac{3}{2} i \sin x = 1$, for real x. [8]

b) Given $f(x) = \frac{x^2 - x}{x - 2}$, obtain all stationary points and classify them using the second derivative test. Obtain all asymptotes and intercepts and hence sketch $f(x)$, explaining your reasoning. [7]

c) Given $f(x)$ as in Q1b, let $g(x) = f(x)[H(x) - H(x-2)]$, where $H(x)$ is the Heaviside function. Sketch $g(x)$, and hence sketch $g_e(x)$ and $g_o(x)$, the even and odd parts of $g(x)$, taking care to label the axes. [5]

2. a) Obtain the integrals:

$$\int \cosh^{-1} x \, dx \,, \qquad \int_0^1 3x^2 \tan^{-1} x \, dx \,, \qquad \int e^{-2x} \sin(3x) \, dx \,.$$

[9]

b) If $f(x) = \ln(1+x)$, show that its Maclaurin series is

$$f(x) \;=\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \, x^n \;=\; x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \,,$$

and find the interval of convergence. [6]

c) Use the first three terms of the series obtained in (b), above, to estimate

$$\int_0^1 \frac{\ln(1+x)}{x} \, dx.$$

Use the remainder term to obtain a bound on the size of the error. [5]

3. a) Find the solution of the first-order ODE

$$\frac{dx}{dt} = \frac{-3t-x}{2x-2t} + \frac{3x}{2t} ,$$

giving $x(t)$ as an explicit function of t. [6]

b) Show that the substitution $x = e^t$ transforms the Euler equation, a second-order linear ODE, with non-constant coefficients:

$$x^2\frac{d^2y}{dx^2} + ax\frac{dy}{dx} + by = f(x) ,$$

into the second-order linear ODE with constant coefficients:

$$\frac{d^2y}{dt^2} + (a-1)\frac{dy}{dt} + by = f(e^t)$$

[7]

c) Given $u(x,y) = x\cosh y$ and $v(x,y) = x\sinh y$, if $f(u,v)$ is any function of u and v, use the chain rule to show that

$$\frac{\partial f}{\partial u} = \cosh y \, \frac{\partial f}{\partial x} - \frac{\sinh y}{x} \, \frac{\partial f}{\partial y} ,$$

and find a similar expression for $\dfrac{\partial f}{\partial v}$. [7]